

Experienced Data Engineer

System Design Case



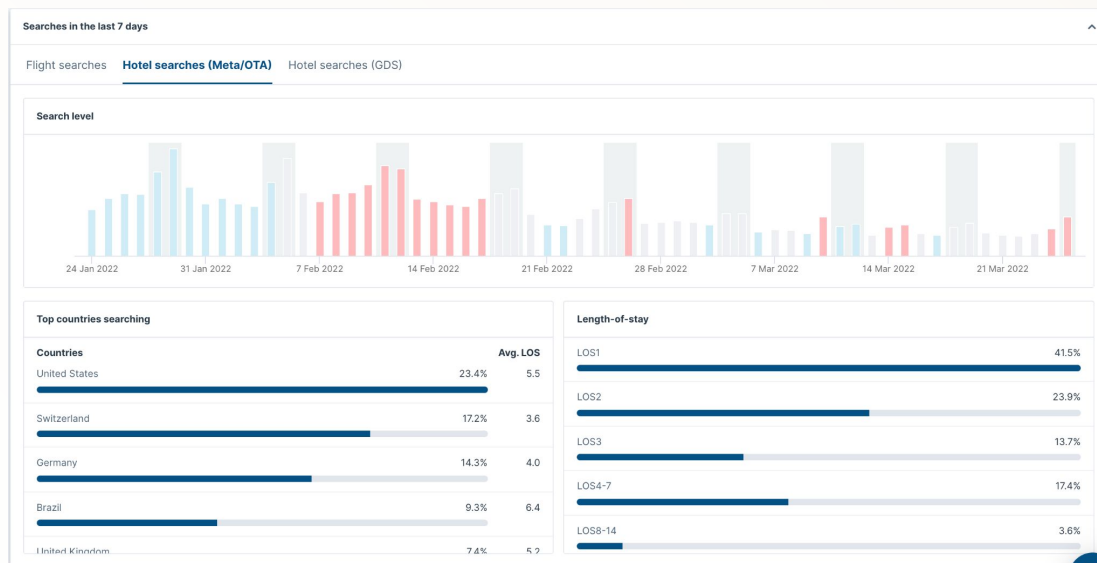
Technical Interview Process

- Preparation:
 - Go through technical case in the next slide and prepare a presentation to explain your solution
 - Let us know if you have specific questions in preparation for the case.
 - Highlight any assumptions you have made while answering the question.
 - Deliverables to be shared 2 days before the interview
- On-site interview:
 - Agenda: 5' intros, 30' case presentation, 30' case questions, 15' other questions, 10' questions for Lighthouse



Technical Case

- We start a partnership with an online travel agency (e.g. booking.com, trivago,..) that will give us insights in how much a hotel is searched for, by whom and for which days
- Goal is to visualise the trends of this data, per city, in our Market Insight product





Technical Case

- Specification of the data feed:
 - Per search done on their platform they will perform an HTTP POST request to our endpoint
 - Volume: 100 req/s
 - Size: 1kB/req
 - JSON data content, example see image

```
{
  "arrival_date": "2022-03-01",
  "departure_date": "2022-03-05",
  "length_of_stay": 4,
  "user_country": "Belgium",
  "hotel_id": 1235,
  "hotel_name": "Holiday Inn Manhattan",
  "hotel_latitude": 40.70825421355257,
  "hotel_longitude": -74.0142071188057,
  "timestamp": "2022-01-24T11:01:12",
  "search_results": [
    {
      "room_name": "Deluxe Suite",
      "price": 124,
      "currency": "USD"
    },
    ...
  ],
  ...
}
```



Technical Case

- Design a system, that is able to receive, store and expose this data to our application.
- The application exposes the following trends per city: popularity of arrival date, country from which the user searched from, and length of stay
- Create an architecture diagram for the pipeline, with a focus on the architecture and technologies you would use. You can use any language or set of technologies you are comfortable with
- Devise and provide a method for validating the incoming data. You can use any language or technology you are comfortable with
- Lay out your strategy for infrastructure provisioning, resource definition, module usage, state management, applying changes, CI/CD deployment strategy
- Give a rough estimation of the monthly costs for the entire pipeline and its infrastructure, explaining any potential cost reduction strategies
- *Consideration:* In your design, consider all aspects that could impact the success and efficiency of the pipeline, including but not limited to data quality & governance, data privacy, system performance, and error handling

Best of luck!

Any questions? Please reach out to your HR contact person.